

Cold run with the second MLS pass — 2026-09-18

The stacked-sphere candidate from the sweep, wired into the real pipeline and run cold on `test6` and the can. Nothing offline this time: both columns in every gallery are what `run.py` wrote to disk.

The change

One extra `mls\_project` call in `\_clean\_cluster` (`pipeline/stages/clean.py`) after the existing 4x pass, limb only, at 16x point spacing. Gated by `MLS\_SECOND\_RADIUS\_MULT` in `pipeline/config.py`, `**default 0 (off)**`, switched on per run with an environment variable:

```
MLS_SECOND_RADIUS_MULT=16 python run.py -i inputs/test6
```

The reference cube is never touched by it, so the scale is identical between a run with the pass and a run without. The config comment carries the sweep that justifies the number.

`test6` — nearly hairless shin, taped every 2 cm

	today (4x)	cold run (4→16)	truth
--- --- --- ---			
limb volume, span	1939.4 **+38.7%**	1703.9 **+21.8%**	1398.6 (tape)
reconstruction	alpha 30x (Poisson $\chi=-22$) **Poisson $\chi=2$** , 112,862 faces		
mesh / points area	1.105	**0.989**	1.00
lower-band girth	20.14 (+6.0%)	19.95 (+5.0%)	19.0
upper-band girth	32.72 (+16.9%)	32.09 (+14.6%)	28.0
cube bbox	11.53 × 11.59 × 10.88	identical	
total time	526 s	123.5 s	

Per slice (`ring\_gallery\_test6\_cold\_run.pdf`):

h	today girth	m/p	cold girth	m/p
0	20.51	1.04	20.67	1.00
8	23.82	1.06	23.82	1.00
14	29.39	1.13	29.14	0.99
20	31.92	1.15	31.54	0.99
26	30.82	1.17	30.82	0.97
28	31.93	1.17	31.85	0.98
mean		1.105		0.989

The solid now sits on the points at every height. The volume drops by $1939.4 / 1703.9 = \textbf{**}1.138\textbf{**}$, against the $\times 1.12$ wrap excess measured on the slices – the fallback's inflation is gone and nothing else moved. The girth came in about 1% smaller, not larger: the merged ring lands slightly inside

where the two sheets were. That is toward the tape, and small.

The can — rigid cylinder, calipers 18.5 girth, 14.5 tall

	today (4×)	cold run (4→16)	truth
--- --- --- ---			
volume	415.5	**387.2**	≤ 394.9 (cylinder bound)
reconstruction	alpha (Poisson $\chi=-12$)	**Poisson $\chi=2$** , 53,514 faces	
mesh / points area	1.003	0.999	
girth	18.45 (+0.9%)	18.45 (+0.9%)	18.5
height	15.48 (+6.8%)	**14.78 (+1.9%)*	14.5
second pass moved	—	median 0.00035, p95 0.0028	

The control held exactly: girth unchanged to the second decimal, and the pass moved half as much as on the limb because a can has almost nothing to merge. The height error — +6.8% before — was the alpha end caps; it is +1.9% now, and the volume sits 2% under the cylinder bound, where a real can with a domed base and tapered neck belongs.

What it does and does not do

****Removes****: the alpha-shape fallback and its wrap, on both objects, through the pipeline's own Poisson and ``_survives_repair``. That was the $\times 1.10$ term.

****Leaves****: the cube-derived scale ($\times 1.133$ on ``test6``; ruler 27.5 cm between bands, pipeline 28.7) and the real girth over-read (+5% ankle, +15% knee). Together they are the +21.8% that remains, and no smoothing reaches either. The caliper on the cube edge is still the single largest lever.

****Offline vs real****: the sweep predicted 1614.5 for ``test6``; the real run gave 1703.9. The per-slice sections agree to 0.001, so the 5% sits in what 2 cm slices do not sample. The real number is the one that counts.

Validation — done 2026-09-19, pass is now the default

test5 (water 2070): 2585.3 / +24.9% / alpha $\chi=-4 \rightarrow 2438.6$ / +17.8% / Poisson.
0_right, 1_left, 6_left (water): Poisson before and after, volumes within 1%, girth within 0.3 cm. Table in EXPERIMENT_LOG.md section 10.
``MLS_SECOND_RADIUS_MULT`` defaults to 16 in config.py.

Validation that was owed before it became the default

- ``test5`` (hairy, noisier shell) and three or four of the 0–6 cohort, with their water truth.
- A cap in centimetres, not just spacing multiples: 16× on a sparser cloud is a larger physical radius.

- The can re-shot on the matte tile, so the control is not a marginal-cube scene (fill 0.815).

Files

```
| file | |
|---|---|
| `ring_gallery_test6_cold_run.pdf` | every 2 cm, today vs cold run, from disk |
| `ring_gallery_can_cold_run.pdf` | the same for the can |
| `ring_gallery_two_runs.py` | draws either from two run directories |
| `output_test6_mls16/`, `output_fanta_orange_mls16/` | the runs (local, gitignored) |
```